ROHAN KUMAR

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PROFESSIONAL SUMMARY

Risk, compliance, and audit analytics professional with hands-on experience at global banks including SMBC, Deutsche Bank, HSBC, and Santander, supporting regulatory examinations, model governance (SR 11-7, CCAR), and control monitoring. Skilled in building SQL/Python data pipelines and reproducible, audit-ready evidence packages that withstand regulatory scrutiny; monitoring control and performance indicators, investigating breaches, and identifying gaps in risk assessments and mitigations. Experienced in explaining and defending data methodologies to regulators, auditors, and cross-functional partners (Engineering, Legal, Product, Risk), and in delivering against tight, inflexible deadlines amid ambiguous, fast-changing requests. Integrates AI/LLM tools (RAG, NLP, prompt engineering) into compliance workflows with responsible-AI controls: human-in-the-loop review, accuracy and hallucination testing, and auditability.

I am currently an F-1 PhD student in Computational Science and Engineering. I am eligible to obtain CPT authorization through my university for a role related to my doctoral program. CPT does not require employer visa sponsorship.
Eligible for change of status F1- H1B (change of status, cap exemption, valid for around 4 years).

PROFESSIONAL EXPERIENCE

Madison-Davis / SMBC Compliance, Jersey City, NJ
Compliance Analytics Consultant | Jun 2025 - Dec 2025

- Developed and enhanced in-house Market Surveillance models to detect potential market manipulation and policy-breach behaviors across Fixed Income and FX asset classes, improving detection precision while reducing false positives by 20%.
- Built analytical workflows using Azure Databricks (Python/SQL) to automate surveillance model inputs and alert reviews, cutting manual processing time on a daily basis by 20-30%.
- Performed data reconciliation, pattern detection, and anomaly analysis using structured and unstructured datasets to strengthen surveillance coverage.
- Evaluated NLP/LLM-assisted approaches for summarizing unstructured communications and supporting alert triage, with human-in-the-loop review and documented controls aligned to SR 11-7.
- Conducted annual threshold tuning and back-testing (BTL) exercises for in-house surveillance models to improve detection sensitivity and reduce false positives.
- Collaborated with Compliance, Technology, and Surveillance SMEs to ensure model logic aligns with SEC, FINRA, and internal SMBC policies.
- Prepared methodology documentation, model change artifacts, and test evidence in alignment with SR 11-7 model risk standards.
- Performed code versioning and model enhancements using GitHub; documented alert logic, UAT/production run procedures in Confluence.
- Produced reproducible, audit-ready evidence packages for surveillance models (population and sample extracts, threshold-tuning and back-testing results, documented methodology) that Internal Audit, Model Risk, and regulatory reviewers could independently re-run and verify.
- Monitored model-performance and alert-quality indicators against defined thresholds, investigated breaches, and documented attestations and remediation actions within required SLAs.
- Identified gaps in surveillance coverage and risk determinations through data reconciliation; partnered with Compliance and Technology to draft action plans and track remediation documentation to closure.
- Explained and defended data methodology, lineage, and model logic to Compliance, Model Risk, and audit stakeholders, building consensus on remediation priorities.

PHD/M.S. Computer Science - Independent Analytics Research, Jersey City, NJ
Self-Directed | Analytics Research & Applied ML Development | Jan 2024 - Present

- Designed and developed a Trade Surveillance Detection System (Python, OOP) integrating real-time market data via Yahoo Finance and CoinGecko APIs; published ML-based anomaly detection research on SSRN (Abstract ID 6576018) covering Fixed Income and crypto market surveillance.
- Building LLM-powered Retrieval-Augmented Generation (RAG) prototypes to query regulatory and model-risk documents (SR 11-7, BCBS 239, SEC/FINRA guidance) with source-grounded, citable answers.

- Applying NLP and LLM techniques (SpaCy, transformer embeddings, prompt engineering) to extract entities, classify, and summarize unstructured financial text for surveillance and compliance cases.
- Researching responsible AI for regulated finance: testing LLM outputs for hallucination, explainability, and auditability, and mapping controls to SR 11-7 model risk governance principles.
- Integrated LLM/AI tools into regulatory document-review workflows to speed research and evidence gathering, applying responsible-AI practices: risk assessment, bias and hallucination testing, and quality and accuracy reviews before outputs are relied upon.
- Continuously building AI skills (prompt and context engineering, LLM evaluation, RAG pipelines) and tracking emerging AI tools for compliance and risk use cases.
- Designing evaluation frameworks for LLM-based assistants (prompt test sets, grounding checks, benchmark comparisons) as part of doctoral research in Computational Science and Engineering.
- Built an end-to-end Customer Support Operations Dashboard in Power BI using Databricks SQL; designed DAX measures, KPI visuals, and executive-level data storytelling layouts.
- Completed advanced coursework in Machine Learning, Data Engineering, and Algorithms as part of M.S. Computer Science program (Monroe University); applied regression modeling and Jupyter-based analysis to real-world datasets.
- Maintained subject matter expertise in financial risk and regulatory compliance through continued engagement with industry literature on BCBS 239, SR 11-7, CCAR/PPNR, and market surveillance frameworks.

Deutsche Bank, New York, NY
Assistant Vice President - Senior Risk Methodology Specialist | Aug 2022 - Dec 2023

- Led strategic analytics for corporate banking portfolios supporting CCAR stress testing and IRRBB, managing financial models with $5B+ in revenue and balance sheet exposures.
- Revamped and calibrated PPNR models, leading to a 15% reduction in forecasting errors and increased model accuracy.
- Supported CCAR stress testing 2023 and MRM validation activities for Corporate Trust, Institutional Cash Management, and Corporate Cash Management portfolios.
- Monitored CCAR/PPNR models and ensured compliance with model risk management standards per SR 11-7 and internal governance policies.
- Reviewed model risk assessments and validation findings for accuracy, confirming determinations captured all relevant risks and mitigations; tracked remediation to closure.
- Prepared supporting data, documentation, and rationale for CCAR regulatory submissions and MRM requests under fixed regulatory deadlines.
- Supported IRRBB analytics by analyzing interest rate sensitivity, balance sheet repricing gaps, and scenario impacts under regulatory stress environments.
- Collaborated with stakeholders across teams to ensure compliance with bank standards and guidelines, resulting in a 30% faster model review cycles, reporting turnaround.

Primesoft Consulting Services (Santander), Jersey City, NJ
Credit Risk Modeler (Contract) | Apr 2020 - Aug 2022

- Developed credit models including risk rating, scorecard, and CCAR loss estimation models leveraging PD/EAD/LGD frameworks, stress testing, and prepayment models.
- Led NDOD (IFRS 9) validation analysis for retail portfolio using SAS, achieving compliance with model development standards.
- Improved data accuracy and efficiency of risk reporting using SQL and SAS for extraction, transformation, and analysis of large, structured datasets.
- Prepared model development documentation for CECL and IFRS 9 compliance, ensuring 98% adherence to established standards.

Huxley / Allegis Global Solution (HSBC), New York, NY
Risk Management Specialist (Contract) | Jun 2018 - Nov 2019

- Managed fraud risk models and related strategies; performed independent model reviews and oversaw vendor model management.
- Executed fraud model monitoring using SAS Enterprise Guide, ensuring 99% compliance with bank standards; conducted annual vendor model reviews leading to 15% improvement in model efficacy.
- Performed independent audits of fraud and vendor models, assessing findings and risk determinations for accuracy and documenting required mitigations.

- Collaborated with IT and model implementation groups for UAT and BRD documentation, achieving 95% accuracy in documentation completeness.

CRISIL / S&P Global, New York, NY
Quantitative Analyst | Sep 2015 - Nov 2017

- Supported CCAR stress testing, time series regression model development, and validation of fraud vendor models and PPNR models.
- Validated models for large investment banks (Citi, BofA, BNP), improving model governance by implementing standardized validation and monitoring processes.
- Communicated validation findings and methodology to client risk and compliance leaders and regulators-facing teams, adapting to changing requests and tight deadlines.
- Developed time series regression models and created SAS datasets, reducing data processing time by 20%.

Kodukulla and Associates Inc., Chicago, IL
Research Analyst | Jan 2015 - May 2015

- Monte Carlo Simulation and DCF analysis Applied to assess project NPV distributions and risk ranges across multiple industries; developed client training materials on quantitative risk methodology, improving estimation accuracy by 10%.

EARLIER CAREER - ENGINEERING & TECHNOLOGY

HCL Technologies - Lead Engineer, Pune, India (May 2012 - Jul 2013)

Tata Motors - Engineer, Pune, India (Jan 2009 - Nov 2011)

Tech Mahindra - Developer/Analyst, Hyderabad, India (Dec 2007 - Dec 2008)

Action Construction Equipment / TAG Design - Design Engineer, India (Sep 2005 - Dec 2007)

EDUCATION

Harrisburg University of Science and Technology, PA – PHD Computational Science and Engineering (Expected 2029)

Monroe University, New Rochelle, NY - M.S. Computer Science, GPA: 4.0

Harrisburg University of Science and Technology, PA - M.S. Analytics, GPA: 3.9

Goldey-Beacom College, Wilmington, DE - MBA, Financial Management, GPA: 3.95

Illinois Institute of Technology, Chicago, IL - M.S., Finance - Risk Management & Corporate Finance

Manipal Institute of Technology, India - B.E., Mechanical Engineering

CERTIFICATIONS

- Databricks Certified Data Engineer Associate
- Equity Essentials, Bloomberg Essentials Training Program, Chicago, IL

TECHNICAL SKILLS

Languages & Analytics: Python, SAS, R, C++, Java, SQL, PostgreSQL, kdb+/q, NoSQL, SQLite

Big Data / Cloud: Azure Databricks, PySpark, Azure, GCP, AWS

ML & Libraries: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, SpaCy

Visualization & BI: Tableau, Power BI, Matplotlib, ggplot

Tools: JIRA, Confluence, MS Project, GitHub, Visual Studio Code, SQL Server

Compliance & Audit: KCI/KRI monitoring, regulatory evidence packages, audit and risk-assessment review, privacy/integrity/model risk, reproducible Jupyter notebooks, SQL population and sample queries

Deep Learning: CNN, RNN, GAN, RAG, Deep Learning Frameworks & Techniques

CORE COMPETENCIES

Trade Compliance & Market Surveillance, Fixed Income & FX Risk, Regulatory Compliance (SEC, FINRA, SR 11-7),
Model Development, Validation & Governance, CCAR Stress Testing, IRRBB, Credit Risk (PD/EAD/LGD),
Predictive Modeling & Machine Learning, Time Series Forecasting, Statistical Analysis,
Stakeholder Management, Cross-Functional Collaboration, Regulatory Documentation